I- Introduction

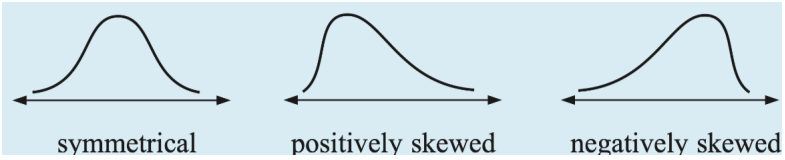
a) Vocabulary

In statistics, we collect information about a group of individuals, then analyse this information to draw conclusions about those individuals.

Definition

Some key terms :

- **Population** – an entire collection of individuals about which we want to draw conclusions.
- **Census** – the collection of information from the whole population.
- **Sample** – a subset of the population which should be chosen at random to avoid bias in the results.
- **Survey** – the collection of information from a sample.
- **Data** – information about individuals in a population.
- **Categorical variable** – describes a particular quality or characteristic which can be divided into categories.
- **Numerical variable** – describes a characteristic which has a numerical value that can be counted or measured.
- **Parameter** – a numerical quantity measuring some aspect of a population.
- **Statistic** – a quantity calculated from data gathered from a sample, usually used to estimate a population parameter.
- **Distribution** – the pattern of variation of data, which may be described as :
  - symmetrical
  - positively skewed
  - negatively skewed



- **Outliers** – data values that are either much larger or much smaller than the general body of data ; they should be included in analysis *unless* they are the result of human or other known error.

b) Numerical variables

Definition

There are two types of numerical variable we will deal with in this course :

A discrete numerical variable takes exact number values and is often a result of counting.

A continuous variable takes numerical values within a certain continuous range. It is usually a result of measuring.

Example

— The height of Year 11 students
— The number of people in a car
— the score out of 20 for a test
— The speed of cars on a stretch of highway

c) Frequency tables

Definition

A tally is used to count the number of 1s, 2s, 3s, and so on. As we read the data from left to right, we place a vertical stroke in the tally column. We use |||| to represent 5.

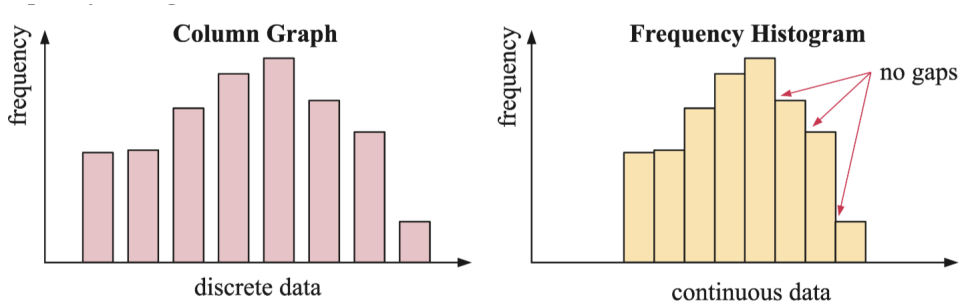
The frequency column summarises the number of each particular data value.

The relative frequency column measures the percentage of the total number of data values that are in each group.

Value	*Tally*	*Frequency* (f)	*% relative frequency*
1	𝍸 |||	8	26.7 ← $\frac{8}{30} \times 100\%$
2	𝍸 ||||	9	30.0
3	𝍸 |	6	20.0
4	𝍸	5	16.7
5	||	2	6.7
	Totals	30	100

Definition

The modal class, or class of values that appears most often, is easy to identify from the highest column of the frequency histogram.



Example

A sample of 20 juvenile lobsters is randomly selected from a tank containing several hundred. Each lobster is measured for length (in cm) and the results are :

4.9, 5.6, 7.2, 6.7, 3.1, 4.6, 6.0, 5.0, 3.7, 7.3,
6.0, 5.4, 4.2, 6.6, 4.7, 5.8, 4.4, 3.6, 4.2, 5.4

a) Organise the data using a frequency table, and hence graph the data.

b) State the modal class for the data.

c) Describe the distribution of the data.

d) Exercises Set I

EXERCISE 1 A frequency table for the heights of a basketball squad is given.

Height (cm)	Frequency
$170 \leqslant H < 175$	1
$175 \leqslant H < 180$	8
$180 \leqslant H < 185$	9
$185 \leqslant H < 190$	11
$190 \leqslant H < 195$	9
$195 \leqslant H < 200$	3
$200 \leqslant H < 205$	3

a) Explain why 'height' is a continuous variable.

b) Construct a frequency histogram for the data. The axes should be clearly marked and labelled, and the graph should have a title.

c) What is the modal class ? Explain what this means.

d) Describe the distribution of the data.

EXERCISE 2 A school has conducted a survey of 60 students to investigate the time it takes for them to travel to school. The following data gives the travel times to the nearest minute :

12, 15, 16, 8, 10, 17, 25, 34, 42, 18, 24, 18, 45, 33, 38,
45, 40, 3, 20, 12, 10, 10, 27, 16, 37, 45, 15, 16,6,2
35, 8, 14, 18, 15, 27, 19, 32, 6, 12, 14, 20, 10, 16, 14,
28, 31, 21, 25, 8, 32, 46, 14, 15, 20, 18, 8, 10, 25, 22

Complete the following :

Time (min)	Tally	Frequency
0 – 9		
10 – 19		
20 – 29		
30 – 39		
40 – 49		

a) Is travel time a discrete or continuous variable ?

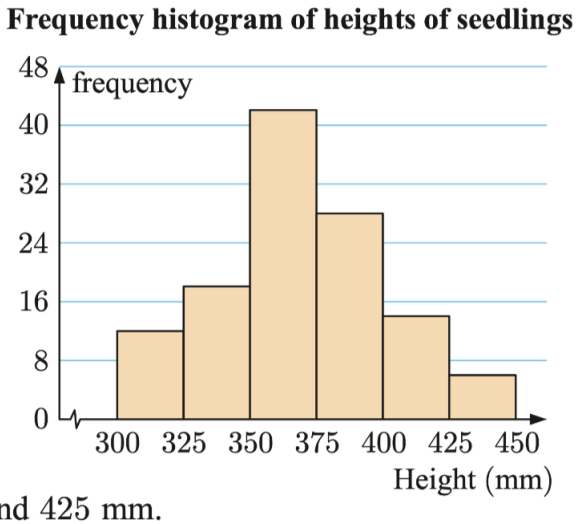
b) Copy and complete the frequency table shown.

c) Describe the distribution of the data.

d) What is the travelling time modal class ?

EXERCISE 3

A plant inspector takes a random sample of six month old seedlings from a nursery and measures their heights to the nearest mm.
The results are shown in the frequency histogram.



a How many of the seedlings are 400 mm or more in height?

b What percentage of the seedlings are between 349 and 400 mm?

c The total number of seedlings in the nursery is 1462. Estimate the number of seedlings which measure:

i less than 400 mm ii between 374 and 425 mm.

II- Centre of data

We can get a better understanding of a data set if we can locate the middle or centre of the data, and also get an indication of its spread or dispersion. Knowing one of these without the other is often of little use.

a) Mode

Definition

For discrete numerical data, the mode is the most frequently occurring value in the data set.

For continuous numerical data, we cannot talk about a mode in this way because no two data values will be exactly equal. Instead we talk about a modal class, which is the class or group that occurs most frequently.

b) Mean

Definition

The mean of a data set is the statistical name for the arithmetic average.

$mean = \frac{\text{sum of all data values}}{\text{the number of data values}}$

Property

Suppose x is a numerical variable. We let :

— x_i be the ith data value

— n be the number of data values in the sample or population

— $\bar{x}$ represent the mean of a sample, so

$\bar{x} = \frac{x_1+x_2+x_3+\ldots+x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$

— μ represent the mean of a population, so

$\mu = \frac{\sum_{i=1}^{n} x_i}{n}$

Remark. In many cases we do not have data from all of the members of a population, so the exact value of μ is often unknown.

Instead we collect data from a sample of the population, and use the mean of the sample $\bar{x}$ as an approximation for μ.

c) Median

Definition

The median is the middle value of an ordered data set.

Remark. An ordered data set is obtained by listing the data from smallest to largest value.

The median splits the data in halves. Half of the data are less than or equal to the median, and half are greater than or equal to it.

Property

For an odd number of data, the median is one of the original data values.

For an even number of data, the median is the average of the two middle values, and hence may not be in the original data set.

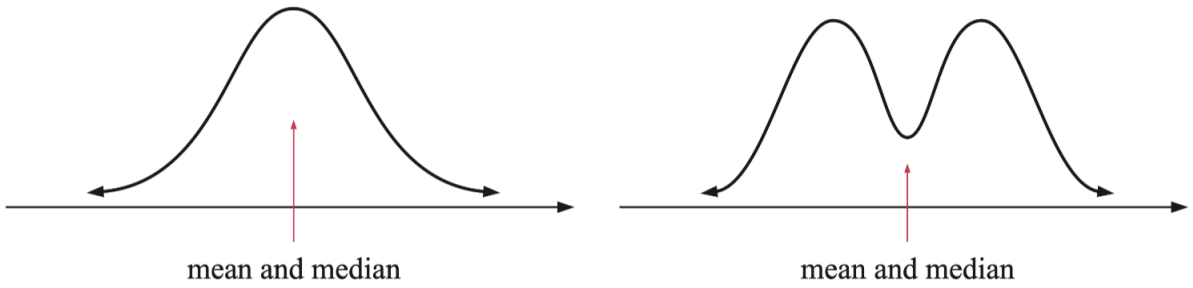
If there are n data values listed in order from smallest to largest, the median is the $\left(\frac{n+1}{2}\right)$th data value.

Example

If $n = 13$, $\frac{13+1}{2} = 7$, so the median is the 7th ordered data value.

If $n = 14$, $\frac{14+1}{2} = 7.5$, so the median is the average of the 7th and 8th ordered data values.

If the data set has symmetry, both the mean and the median should accurately measure the centre of the distribution.



Example

The number of faulty products returned to an electrical goods store over a 21 day period is :

3 4 4 9 8 8 6 4 7 9 1 3 5 3 5 9 8 6 3 7 1

For this data set, find :

1. the mean
2. the median
3. the mode.

Example

If 6 people have a mean mass of 53.7 kg, find their total mass.

Example

If 6 people have a mean mass of 53.7 kg, find their total mass.

Property

Mean using frequencies

$\bar{x} = \frac{f_1x_1+f_2x_2+f_3x_3+\cdots+f_kx_k}{f_1+f_2+f_3+\cdots+f_k}$ where k is the number of different data values.

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i}$$

This formula is often abbreviated as $\bar{x} = \frac{\sum fx}{\sum f}$.

Example

The table below shows the number of aces served by tennis players in their first sets of a tournament.

Number of aces	1	2	3	4	5	6
Frequency	4	11	18	13	7	2

Determine the :

1. mean
2. median
3. mode

for this data.

When information has been gathered in groups or classes, we use the midpoint or mid-interval value to represent all scores within that interval.

Example

Estimate the mean of the following *ages of bus drivers* data, to the nearest year :

Age (yrs)	21–25	26–30	31–35	36–40	41–45	46–50	51–55
Frequency	11	14	32	27	29	17	7

d) Exercises Set II

EXERCISE 4

The annual salaries of ten office workers are:

$23 000, $46 000, $23 000, $38 000, $24 000,
$23 000, $23 000, $38 000, $23 000, $32 000

a Find the mean, median, and modal salaries of this group.

b Explain why the mode is an unsatisfactory measure of the middle in this case.

c Is the median a satisfactory measure of the middle of this data set?

EXERCISE 5

The selling prices of the last 10 houses sold in a certain district were as follows:

$146 400, $127 600, $211 000, $192 500,
$256 400, $132 400, $148 000, $129 500,
$131 400, $162 500

a Calculate the mean and median selling prices and comment on the results.

b Which measure would you use if you were:

i a vendor wanting to sell your house

ii looking to buy a house in the district?

EXERCISE 6

A sample of 10 measurements has a mean of 15.7 and a sample of 20 measurements has a mean of 14.3. Find the mean of all 30 measurements.

EXERCISE 7

The following frequency table records the number of phone calls made in a day by 50 fifteen-year-olds.

Number of phone calls	*Frequency*
0	5
1	8
2	13
3	8
4	6
5	3
6	3
7	2
8	1
11	1

a For this data, find the:
i mean **ii** median **iii** mode.

b Construct a column graph for the data and show the position of the mean, median, and mode on the horizontal axis.

c Describe the distribution of the data.

d Why is the mean larger than the median for this data?

e Which measure of centre would be the most suitable for this data set?

EXERCISE 8

Families at a school in Australia were surveyed, and the number of children in each family recorded. The results of the survey are shown alongside.

Number of children	*Frequency*
1	5
2	28
3	15
4	8
5	2
6	1
Total	59

a Calculate the:
i mean **ii** mode **iii** median.

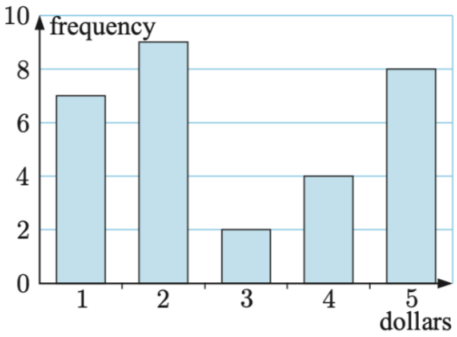
b The average Australian family has 2.2 children. How does this school compare to the national average?

c The data set is skewed. Is the skewness positive or negative?

d How has the skewness of the data affected the measures of the centre of the data set?

EXERCISE 9

The frequency column graph gives the value of donations for the Heart Foundation collected in a particular street.



a Construct a frequency table from the graph.

b Determine the total number of donations.

c Find the: **i** mean **ii** median **iii** mode of the donations.

d Which of the measures of central tendency can be found easily using the graph only?

EXERCISE 10

The table shows the petrol sales in one day by a number of city service stations.

Litres (L)	*Frequency*
$2000 < L \leqslant 3000$	4
$3000 < L \leqslant 4000$	4
$4000 < L \leqslant 5000$	9
$5000 < L \leqslant 6000$	14
$6000 < L \leqslant 7000$	23
$7000 < L \leqslant 8000$	16

a How many service stations were involved in the survey?

b Estimate the total amount of petrol sold for the day by the service stations.

c Find the approximate mean sales of petrol for the day.

EXERCISE 11

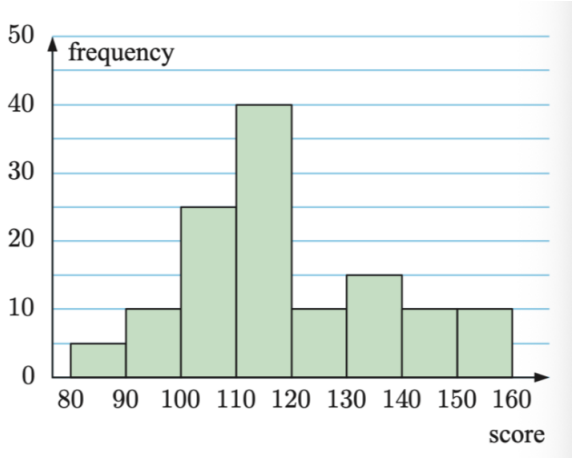
This frequency histogram illustrates the results of an aptitude test given to a group of people seeking positions in a company.

a How many people sat for the test?

b Estimate the mean score for the test.

c What fraction of the people scored less than 100 for the test?

d What percentage of the people scored more than 130 for the test?



III- Spread of data

a) Range

Definition

The range is the difference between the maximum (largest) and the minimum (smallest) data value.

range = maximum − minimum

Example

A library surveys 20 borrowers each day from Monday to Friday, and records the number who are not satisfied with the range of reading material. The results are : 3 7 6 8 11.

The following year the library receives a grant that enables the purchase of a large number of books. The survey is then repeated and the results are : 2 3 5 4 6.

Find the range of data in each survey.

Remark. Is the range reliable to measure the spread of data ?

b) Quartiles

Definition

The middle value of the lower half is called the lower quartile or 25th percentile. One quarter or 25% of the data have values less than or equal to the lower quartile. 75% of the data have values greater than or equal to the lower quartile.

Definition

The middle value of the upper half is called the upper quartile or 75th percentile. One quarter or 25% of the data have values greater than or equal to the upper quartile. 75% of the data have values less than or equal to the upper quartile.

Definition

The interquartile range is the range of the middle half or 50% of the data.

Property

IQR = $Q_3 - Q_1$.
interquartile range = upper quartile − lower quartile

Example

For the data set : 7, 3, 1, 7, 6, 9, 3, 8, 5, 8, 6, 3, 7, 1, 9 find the :

1. median
2. lower quartile
3. upper quartile
4. interquartile range

Example

For the data set : 6, 4, 9, 15, 5, 13, 7, 12, 8, 10, 4, 1, 13, 1, 6, 4, 5, 2, 8, 2 find :

1. the median
2. Q_1
3. Q_3
4. the interquartile range

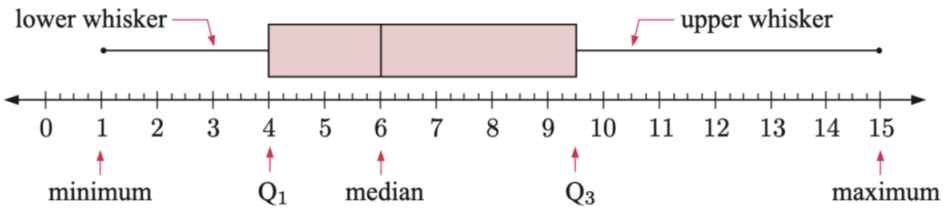
c) Boxplots

Definition

A boxplot or box and whisker plot is a visual display of some of the descriptive statistics of a data set. It shows :

— the minimum value
— the lower quartile (Q_1)
— the median (Q_2)
— the upper quartile (Q_3)
— the maximum value

These five numbers form the five-number summary of the data set.



Example

Consider the data set : 8, 2, 3, 9, 6, 5, 3, 2, 2, 6, 2, 5, 4, 5, 5, 6

1. Construct the five-number summary for this data.
2. Draw a boxplot.
3. Find the :
   (a) range
   (b) interquartile range of the data.
4. Find the percentage of data values less than 3.

d) Outliers

Definition

Outliers are extraordinary data that are separated from the main body of the data. Outliers are either much larger or much smaller than most of the data.

A commonly used test to identify outliers involves the calculation of upper and lower boundaries :

— The upper boundary $=$ upper quartile $+1.5 \times$ IQR.
Any data larger than the upper boundary is an outlier.

— The lower boundary $=$ lower quartile $-1.5 \times$ IQR.
Any data smaller than the lower boundary is an outlier.

Remark. Outliers are marked with an asterisk on a boxplot. There may be more than one outlier at either end.

Example

Test the following data for outliers and hence construct a boxplot for the data : 3, 7, 8, 8, 5, 9, 10, 12, 14, 7, 1, 3, 8, 16, 8, 6, 9, 10, 13, 7

d) Exercises Set III

EXERCISE 12

The heights of 20 ten year olds were recorded (in cm) as follows:

109	111	113	114	114	118	119	122	122	124
124	126	128	129	129	131	132	135	138	138

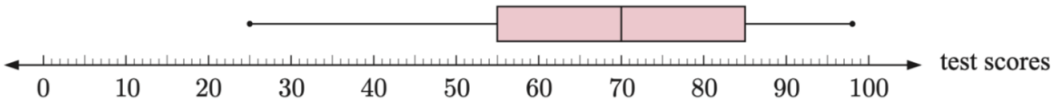
a Find the:
- **i** median height
- **ii** upper and lower quartiles of the data.

b Copy and complete the following statements:
- **i** "Half of the children are no more than ..... cm tall."
- **ii** "75% of the children are no more than ..... cm tall."

c Find the: **i** range **ii** interquartile range for the height of the ten year olds.

d Copy and complete: "The middle 50% of the children have heights spread over ..... cm."

EXERCISE 13



The boxplot summarises the class results for a test out of 100 marks.

a Copy and complete the following statements about the test results:
- **i** The highest mark scored for the test was ...., and the lowest mark was ....
- **ii** Half of the class scored a mark greater than or equal to ....
- **iii** The top 25% of the class scored at least .... marks for the test.
- **iv** The middle half of the class had scores between .... and .... for this test.

b Find the range of the data set.

c Find the interquartile range of the data set.

EXERCISE 14

James goes bird watching for 25 days. The number of birds he sees each day are:

12, 5, 13, 16, 8, 10, 12, 18, 9, 11, 14, 14, 22, 9, 10, 7, 9, 11, 13, 7, 10, 6, 13, 3, 8.

a Find the median, lower quartile, and upper quartile of the data set.

b Find the interquartile range of the data set.

c Find the lower and upper boundaries for outliers.

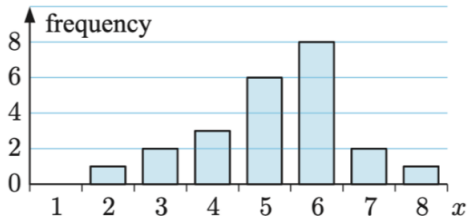
d Are there any outliers?

e Draw a boxplot of the data set.

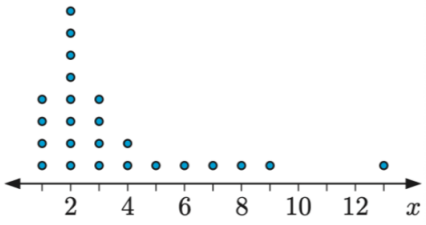
EXERCISE 15

Match the graphs **A**, **B**, **C**, and **D** to the boxplots **I**, **II**, **III**, and **IV**.

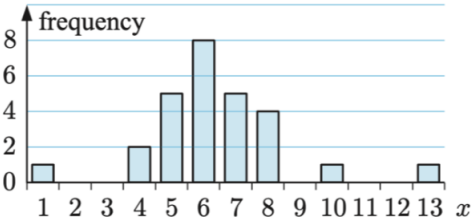
A



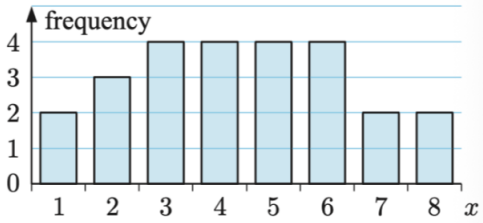
B



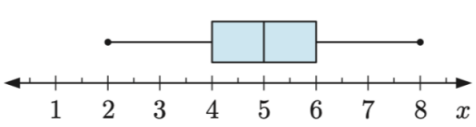
C



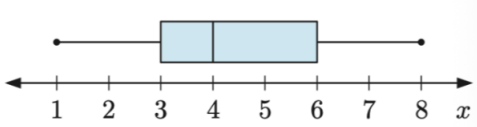
D



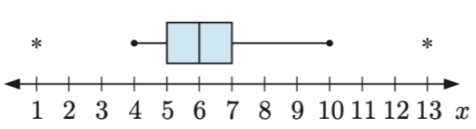
I



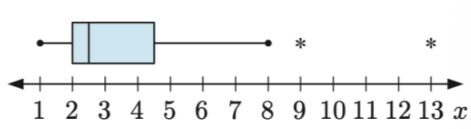
II



III



IV



IV- Cumulative frequency graphs

Definition

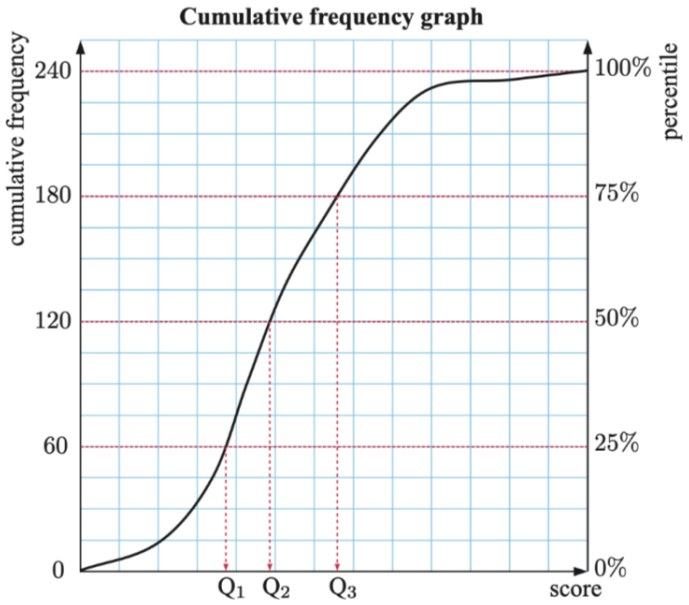
A percentile is the score which a certain percentage of the data lies at or below.

Remark. — the lower quartile (Q_1) is the 25th percentile

— the median (Q_2) is the 50th percentile

— the upper quartile (Q_3) is the 75th percentile.

Another way to calculate percentiles is to add a separate scale to a cumulative frequency graph. On the graph alongside, the cumulative frequency is read from the axis on the left side, and each value corresponds to a percentile on the right side.



EXERCISE 16

The examination scores of a group of students are shown in the table.

Draw a cumulative frequency graph for the data and use it to answer the following:

a Find the median examination mark.

b How many students scored 65 marks or less?

c How many students scored at least 50 but less than 70 marks?

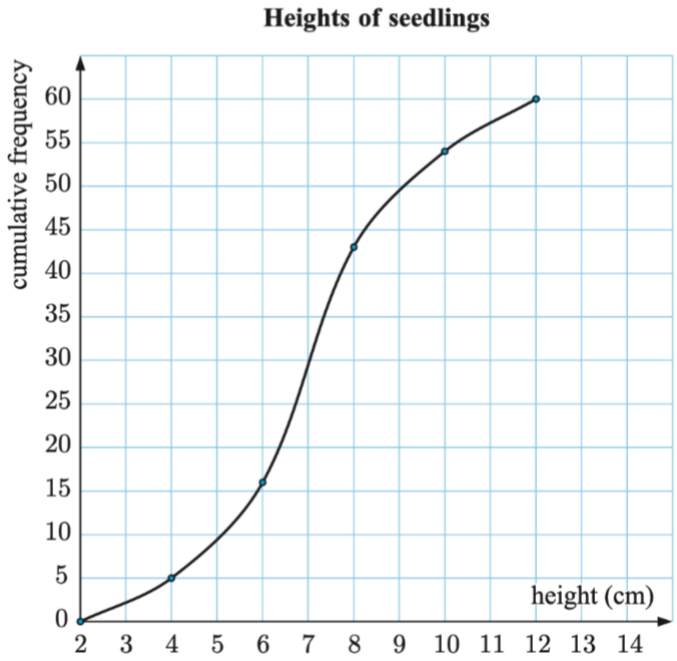
d If the pass mark was 45, how many students failed?

e If the top 16% of students were awarded credits, what was the credit mark?

Score	*Frequency*
$10 \leqslant x < 20$	2
$20 \leqslant x < 30$	5
$30 \leqslant x < 40$	7
$40 \leqslant x < 50$	21
$50 \leqslant x < 60$	36
$60 \leqslant x < 70$	40
$70 \leqslant x < 80$	27
$80 \leqslant x < 90$	9
$90 \leqslant x < 100$	3

EXERCISE 17

A botanist has measured the heights of 60 seedlings and has presented her findings on the cumulative frequency graph below.



a How many seedlings have heights of 5 cm or less?

b What percentage of seedlings are taller than 8 cm?

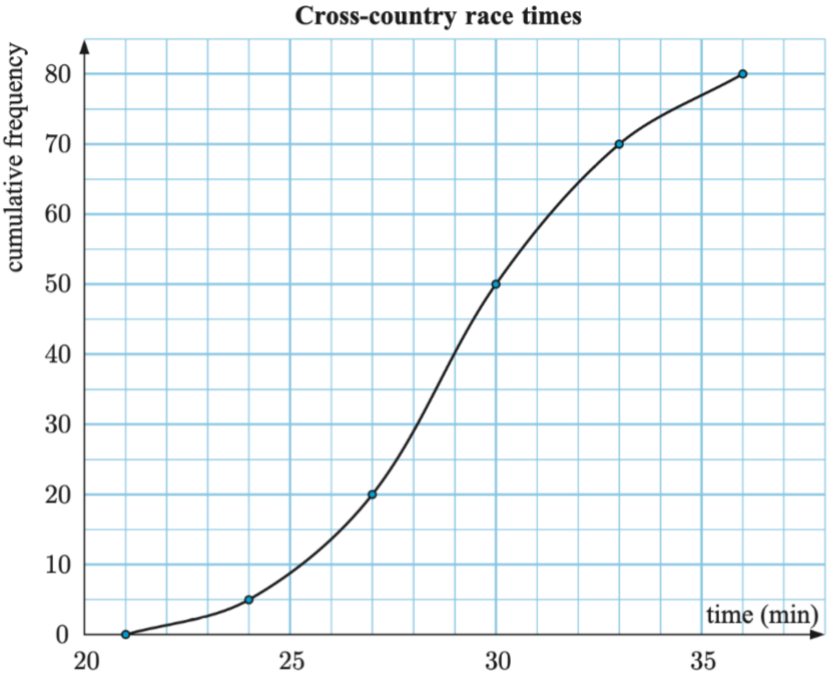
c Find the median height.

d Find the interquartile range for the heights.

e Find the 90th percentile for the data and explain what your answer means.

EXERCISE 18

The following cumulative frequency graph displays the performance of 80 competitors in a cross-country race.



a Find the lower quartile time.

b Find the median.

c Find the upper quartile.

d Find the interquartile range.

e Estimate the 40th percentile.

f Use the cumulative frequency curve to complete the following table:

Time t (mins)	$21 \leqslant t < 24$	$24 \leqslant t < 27$	$27 \leqslant t < 30$	$30 \leqslant t < 33$	$33 \leqslant t < 36$
Number of competitors					

V- Variance and Standard deviation

The problem with using the range and theIQR as measures of spread or dispersion of scores is that both of them only use two values in their calculation.

Definition

The variance of a sample of n data values is

$$s_n^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}.$$

Remark. — $(x_i - \bar{x})^2$ is also a measure of how far x_i deviates from $\bar{x}$.

— If $\sum_{i=1}^{n}(x_i - \bar{x})^2$ is small, it will indicate that most of the data values are close to $\bar{x}$.

— Dividing by n gives an indication of how far, on average, the data is from the mean.

Definition

For a data set of n values, $s_n = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}}$ is called the standard deviation.

Remark. The standard deviation is a non-resistant measure of spread. This is due to its dependence on the mean of the sample and because extreme data values will give large values for $(x_i - \bar{x})^2$.

It is only a useful measure if the distribution is approximately symmetrical.

The IQR and percentiles are more appropriate tools for measuring spread if the distribution is considerably skewed.

Example

A library surveys 20 borrowers each day from Monday to Friday, and records the number who are not satisfied with the range of reading material. The results are :

3 7 6 8 11.

The following year the library receives a grant that enables the purchase of a large number of books. The survey is then repeated and the results are :

2 3 5 4 6.

1. Find the mean and standard deviation for each survey using :
   (a) the formula
   (b) technology.
2. What do these statistics tell us ?

Given statistics from a sample, we can make inferences about the population.

To help distinguish between a sample and the whole population, we use different notation for the mean, variance, and standard deviation. This is shown in the table opposite.

In general, the population mean μ and standard deviation σ will be unknown.

	sample	*population*
mean	$\overline{x}$	μ
variance	s_n^2	σ^2
standard deviation	s_n	σ

When a sample of size n is used to draw inference about a population :

— the mean $\bar{x}$ of the sample is an unbiased estimate of μ

— s_n is an estimate of the standard deviation σ.

Property

Standard deviation for grouped data :

Suppose there are n scores in a set of grouped data with mean $\bar{x}$. If the different data values are $x_1, x_2, \ldots, x_k$, with corresponding frequencies $f_1, f_2, \ldots, f_k$, then the standard deviation is given by

$$s_n = \sqrt{\frac{\sum f_i(x_i - \bar{x})^2}{n}}.$$

Example

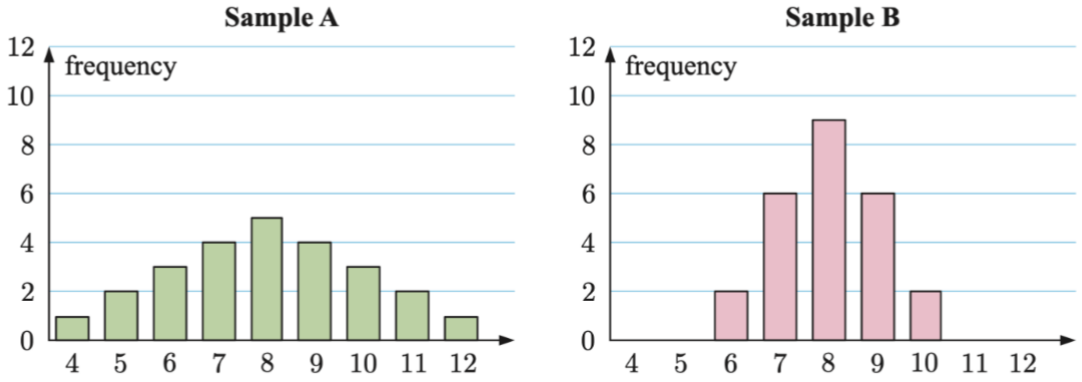
Find the standard deviation of the distribution using :

1. the standard deviation formula
2. technology.

Score	1	2	3	4	5
Frequency	1	2	4	2	1

EXERCISE 19

The column graphs show two distributions:



a By looking at the graphs, which distribution appears to have wider spread?

b Find the mean of each sample.

c Find the standard deviation for each sample. Comment on your answers.

d The other measures of spread for the two samples are given alongside:
In what way does the standard deviation give a better description of how the data is distributed?

Sample	*Range*	*IQR*
A	8	3
B	4	2

EXERCISE 20

The weights in kg of seven footballers are: 79, 64, 59, 71, 68, 68, 74.

a Find the mean and standard deviation for this group.

b When measured five years later, each footballer's weight had increased by exactly 10 kg. Find the new mean and standard deviation.

c Comment on your results in general terms.

EXERCISE 21

The following table shows the decrease in cholesterol levels in 6 volunteers after a two week trial of special diet and exercise.

Volunteer	A	B	C	D	E	F
Decrease in cholesterol	0.8	0.6	0.7	0.8	0.4	2.8

a Find the standard deviation of the data.

b Recalculate the standard deviation with the outlier removed.

c Discuss the effect of an extreme value on the standard deviation.

EXERCISE 22

The number of toothpicks in a random sample of 48 boxes was counted and the results tabulated.

Number of toothpicks	33	35	36	37	38	39	40
Frequency	1	5	7	13	12	8	2

a Find the mean and standard deviation of the number of toothpicks in the boxes.

b Estimate the mean and standard deviation of the population from which the sample was taken.

EXERCISE 23

The lengths of 30 randomly selected 12-day old babies were measured and the following data obtained:

a Estimate the mean length and the standard deviation of the lengths.

b Estimate the mean and standard deviation of the population from which the sample was taken.

Length (cm)	*Frequency*
$40 \leqslant L < 42$	1
$42 \leqslant L < 44$	1
$44 \leqslant L < 46$	3
$46 \leqslant L < 48$	7
$48 \leqslant L < 50$	11
$50 \leqslant L < 52$	5
$52 \leqslant L < 54$	2

EXERCISE 24

The weekly wages (in dollars) of 200 randomly selected steel workers are given alongside:

a Estimate the mean and the standard deviation of the wages.

b Estimate the mean and standard deviation of the population from which the sample was taken.

Wage ($)	*Number of workers*
360 - 369.99	17
370 - 379.99	38
380 - 389.99	47
390 - 399.99	57
400 - 409.99	18
410 - 419.99	10
420 - 429.99	10
430 - 439.99	3